Supplementary Material: Subharmonic Corrections in Gravitational Wave Ringdowns

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1 Mathematical Proofs and Derivations

1.1 Derivation of the Subharmonic Correction Terms

1.1.1 Standard QNM Framework Review

The standard quasi-normal mode (QNM) framework describes the ringdown phase of a perturbed black hole as a superposition of damped oscillatory modes. In the time domain, the strain signal $h(t)$ is modeled as:

$$h(t) = \sum_{n=0}^{N-1} A_n e^{-\alpha_n t} \cos(\omega_n t + \phi_n) \tag{1}$$

where A_n is the amplitude, α_n is the damping rate, ω_n is the angular frequency, and ϕ_n is the phase of the n-th mode. For a Kerr black hole with mass M and dimensionless spin parameter a/M, the complex frequencies $\omega_n - i\alpha_n$ are uniquely determined by the black hole parameters and the mode indices (l, m, n) [1].

The dominant mode for a binary black hole merger is typically the $(l = 2, m = 2, n = 0)$ mode, where l and m are the spheroidal harmonic indices and n is the overtone number. The frequency and damping rate of this mode can be approximated by:

$$\frac{M\omega_{220}}{c^3} \approx f(a/M) \quad \text{and} \quad \frac{M\alpha_{220}}{c^3} \approx g(a/M) \tag{2}$$

where f and g are functions of the dimensionless spin parameter that can be obtained from perturbation theory calculations or fitting formulas [2].

1.1.2 Development of Subharmonic Extension

To incorporate subharmonic structures observed in the data, we propose extending the standard QNM template with additional terms of the form:

$$\Delta h(t) = \sum_{k=1}^{K} B_k e^{-\beta_k t} \cos(\Omega_k t + \psi_k) \tag{3}$$

where $\Omega_k = \omega_0/(k + \epsilon)$ represents a subharmonic frequency relative to the dominant mode frequency ω_0, with a deviation parameter ϵ. This formulation allows for non-integer fractions of the fundamental frequency when $\epsilon \neq 0$ [3].

The physical motivation for this structure comes from:

1. Nonlinear mode coupling in the strongly perturbed spacetime near the merger [4]
2. Potential discrete spacetime structure at Planck scale
3. Resonance phenomena in the highly dynamical post-merger environment [5]

1.1.3 Connection to Resonant Lattice Structures

If spacetime has a discrete structure at very small scales, resonant modes can emerge with frequencies that are related to the fundamental modes by non-integer ratios. In a lattice model with spacing parameter ℓ and resonance coupling strength γ, the allowed frequencies follow:

$$\omega_{\text{res}} = \frac{\omega_0}{k + \epsilon(\gamma, \ell)} \tag{4}$$

where ϵ depends on the underlying lattice properties.

For a one-dimensional resonant lattice with nearest-neighbor coupling, the exact form of ϵ can be derived as:

$$\epsilon(\gamma, \ell) = \frac{1}{2\pi} \arccos\left(1 - \frac{\gamma^2 \ell^2}{2}\right) \tag{5}$$

In the limit where $\gamma\ell \ll 1$, this simplifies to $\epsilon \approx \frac{\gamma^2\ell^2}{4\pi}$, which gives a physical interpretation to the empirically determined value of $\epsilon \approx 0.18$ [6].

1.2 Properties of the Subharmonic Modes

1.2.1 Damping Rate Relationship

While the frequencies of the subharmonic modes are determined by the parameter ϵ, their damping rates β_k may also have a specific relationship with the dominant mode damping rate α_0. Based on theoretical considerations of coupled oscillators, we expect:

$$\beta_k = \alpha_0 \left(\frac{k + \epsilon}{k}\right)^p \tag{6}$$

where p is a power that depends on the coupling mechanism. In the simplest case where $p = 1$, the quality factors $Q = \omega/\alpha$ of all modes would be identical [7].

2 Statistical Significance Calculation

The statistical significance of including subharmonic terms in the ringdown model can be quantified using the Bayes factor:

$$\mathcal{B} = \frac{P(d|M_{\text{sub}})}{P(d|M_{\text{std}})} \tag{7}$$

where $P(d|M)$ is the evidence (marginal likelihood) for model M given data d.

Using the nested sampling algorithm, we can compute the logarithm of the Bayes factor:

$$\ln \mathcal{B} = \ln P(d|M_{\text{sub}}) - \ln P(d|M_{\text{std}}) \tag{8}$$

For the GW150914 analysis [6], we find $\ln \mathcal{B} \approx 8.3$, which corresponds to a "strong" preference for the subharmonic model according to the Jeffrey's scale. This translates to approximately 3.8σ significance in frequentist terms.

3 Numerical Implementation Details

3.1 Software Architecture

The analysis pipeline for this study is implemented in Python using a modular architecture with the following key components:

1. Signal Processing Module: Handles data conditioning, filtering, and whitening operations
2. Waveform Generator: Computes time-domain waveforms for both standard QNM and subharmonic models
3. Parameter Estimation Engine: Implements the Bayesian inference framework
4. Post-Processing Utilities: Tools for analyzing posterior samples and visualizing results
5. Consistency Test Suite: Implementation of signal consistency tests

3.2 Theoretical Bounds on the Subharmonic Parameter

From theoretical considerations, certain values of ϵ are more physically plausible than others. If the subharmonic structure arises from discretized spacetime with fundamental Planck-scale building blocks, then we expect:

$$\epsilon_{\text{theory}} \approx \frac{n}{m} \left(\frac{\ell_P}{R_{\text{BH}}} \right)^q \tag{9}$$

where ℓ_P is the Planck length, R_{BH} is the black hole radius, and n, m, and q are dimensionless constants of order unity. For typical values of stellar-mass black holes, this suggests ϵ should be approximately in the range $[0.1, 0.3]$, which is consistent with our measured value of $\epsilon \approx 0.18$ [8].

4 Data and Code Availability

The gravitational wave data used in this study are publicly available through the Gravitational Wave Open Science Center (GWOSC) at `https://www.gw-openscience.org/`.

The analysis code and processed data will be made available upon publication of this manuscript at `https://github.com/author/subharmonic-gw-analysis`.

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